

Lab program-37:

37. Construct a C program to simulate the First Come First Served disk scheduling algorithm.

Code:

```
#include <stdio.h>
#include <stdlib.h>

int main(void) {
    int n, head, seek_time = 0;

    printf("Enter the number of disk requests: ");
    scanf("%d", &n);
    if (n <= 0) return 0;

    int *request_queue = malloc((size_t)n * sizeof(int));
    if (request_queue == NULL) return 1;

    printf("Enter the disk request queue:\n");
    for (int i = 0; i < n; i++)
        scanf("%d", &request_queue[i]);

    printf("Enter the initial position of the disk head: ");
    scanf("%d", &head);

    printf("\nFCFS Disk Scheduling:\n");
    printf("Head Movement Sequence: %d", head);
    for (int i = 0; i < n; i++) {
        seek_time += abs(head - request_queue[i]);
        head = request_queue[i];
        printf(" -> %d", head);
    }
}
```

```

    printf("\nTotal Seek Time: %d\n", seek_time);
    printf("Average Seek Time: %.2f\n", (float)seek_time / n);

    free(request_queue);
    return 0;
}

```

OUTPUT:

```

C:\Users\jeeva\OneDrive\Doc... x
Enter the number of disk requests: 3
Enter the disk request queue:
222
22
123
Enter the initial position of the disk head: 1

FCFS Disk Scheduling:
Head Movement Sequence: 1 -> 222 -> 22 -> 123
Total Seek Time: 522
Average Seek Time: 174.00

```